EXPERIMENT 9

AIM: **Create a TODO Application in ReactJS and Deploy it on GitHub**

SOLUTION:

A TODO application allows users to add, display, delete and edit tasks.

Procedure

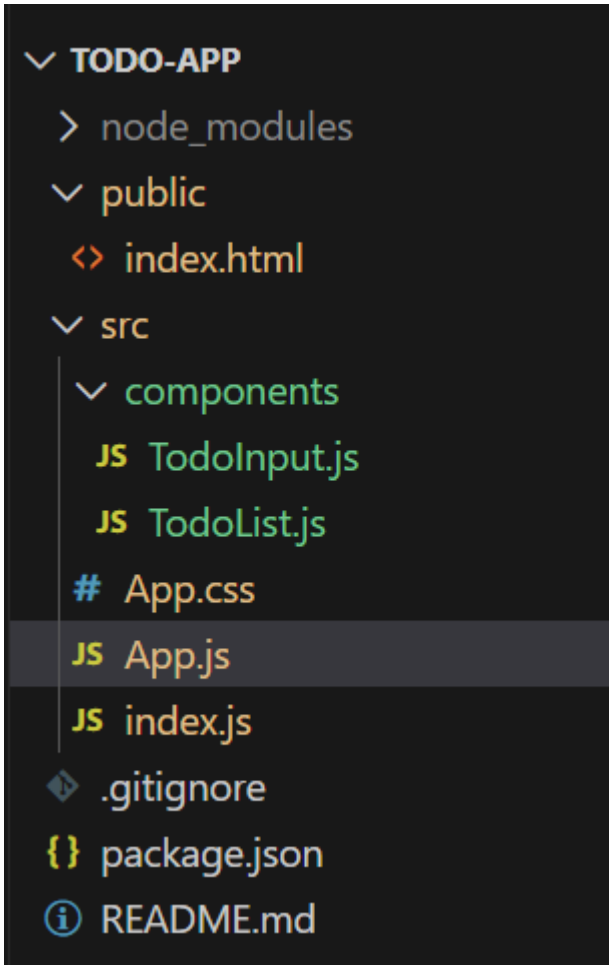
Step 1: Create React Application

npx create-react-app todo-app

Step 2: Navigate to Project Folder

cd todo-app

todo-app Folder Structure:



Program:

public/index.html

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>React App</title>
  </head>
  <body>
```

```
    <div id="root"></div>
  </body>
</html>
```

src/components/TodoInput.js

```
import { useState } from 'react';
function TodoInput({ addTodo }) {
  const [input, setInput] = useState("");

  const handleAdd = () => {
    addTodo(input);
    setInput("");
  };
  return (
    <div className="input-section">
      <input
        type="text"
        placeholder="add task..."
        value={input}
        onChange={(e) => setInput(e.target.value)}
      />
      <button onClick={handleAdd}>ADD</button>
    </div>
  );
}
export default TodoInput;
```

src/components/TodoList.js

```
import React from 'react'
function TodoList({ todos, deleteTodo, editTodo }) {
  const handleEdit = (index, oldText) => {
    const newText = prompt("Edit task:", oldText);
    if (newText !== null) {
      editTodo(index, newText);
    }
  };
```

```
  return (
    <div className="list-section">
      {todos.map((todo, index) => (
        <div key={index} className="todo-item">
          <span>{todo}</span>
          <div>
            <button onClick={() => deleteTodo(index)}>Delete</button>
            <button onClick={() => handleEdit(index, todo)}>Edit</button>
          </div>
        </div>
      ))}
    </div>
  );
}
export default TodoList
```

src/index.js

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```

src/App.js

```
import './App.css';
import { useState } from 'react';
import TodoInput from './components/TodoInput';
import TodoList from './components/TodoList';

function App() {
  const [todos, setTodos] = useState([]);

  const addTodo = (text) => {
```

```
    if (text === "") return;
    setTodos([...todos, text]);
  };
  const deleteTodo = (index) => {
    const newTodos = todos.filter((todo, i) => i !== index);
    setTodos(newTodos);
  };
  const editTodo = (index, newText) => {
    const updated = [...todos];
    updated[index] = newText;
    setTodos(updated);
  };
  return (
    <div className="container">
      <h1>TODO LIST</h1>
      <TodoInput addTodo={addTodo} />
      <TodoList
        todos={todos}
        deleteTodo={deleteTodo}
        editTodo={editTodo}
      />
    </div>
  );
}
export default App;
```

src/App.css

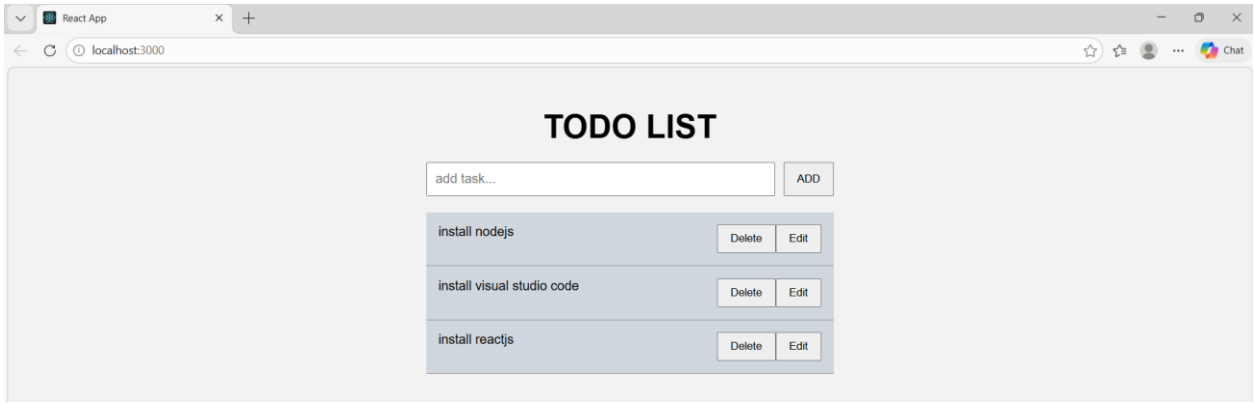
```
body {
  background-color: #f2f2f2;
  font-family: Arial, sans-serif;
}
.container {
  width: 500px;
  margin: 50px auto;
}

h1 {
```

```
  text-align: center;
  font-size: 40px;
  margin-bottom: 20px;
}
.input-section {
  display: flex;
  gap: 10px;
  margin-bottom: 20px;
}
input {
  flex: 1;
  padding: 10px;
  font-size: 16px;
}
button {
  padding: 8px 15px;
  cursor: pointer;
}
.list-section {
  background-color: #cfd6dd;
}
.todo-item {
  display: flex;
  justify-content: space-between;
  padding: 15px;
  border-bottom: 1px solid #999;
}
```

Step 3: Start Application

```
>npm start
```

Output:


React App
localhost:3000
Chat
TODO LIST
add task...
ADD
install nodejs
Delete
Edit
install visual studio code
Delete
Edit
install reactjs
Delete
Edit